

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 3.6.A** — Identify the amplitude, vertical shift (midline), period, and phase shift of a sinusoidal function from its equation. **Identify $|a|$, d , $2\pi/|b|$, and $-c$ from $a \sin(b(\theta + c)) + d$.**
- LO 3.6.A** — Explain how each parameter in $f(\theta) = a \sin(b(\theta + c)) + d$ corresponds to an additive or multiplicative transformation of $y = \sin \theta$. **a dilates vertically; d shifts vertically; b dilates horizontally; c shifts horizontally.**

Vocabulary & Key Concepts

sinusoidal function: A function equivalent to $a \sin(b(\theta + c)) + d$ (or the cosine form) with $a \neq 0$. A **transformation** of the parent sine or cosine function.

amplitude: Equals $|a|$ — half the vertical distance from the **midline** to the **maximum (or minimum)**. Controls the vertical **stretch or compression**.

midline: The horizontal line $y = d$. The vertical **shift** of the function. Average of max and min outputs.

period: Length of one complete cycle; $T = \frac{2\pi}{|b|}$. Controlled by the horizontal dilation factor b .

phase shift: Horizontal translation equal to $-c$ when the function is written in factored form $b(\theta + c)$. Positive c shifts **left**; negative c shifts **right**.

Hook

Entry Question: (Graph as in blank notes.)

Q1. Graph A has amplitude **1** and midline $y = 0$.

Q2. Graph B has amplitude **3** and midline $y = 1$.

Q3. What equation could describe Graph B? $y = 3 \sin \theta + 1$

LO 3.6.A — Part 1: Vertical Transformations (a and d) — Key

Vertical Dilation: $g(\theta) = a \sin(\theta)$

Multiplying all outputs by a changes the **amplitude**. The new amplitude is $|a|$. If $a < 0$, the graph is also **reflected (flipped)** over the midline.

Function	Amplitude	Max	Min	Midline
$y = \sin \theta$	1	1	-1	$y = 0$
$y = 4 \sin \theta$	4	4	-4	$y = 0$
$y = -2 \sin \theta$	2	2	-2	$y = 0$
$y = \frac{1}{2} \sin \theta$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$y = 0$

Vertical Translation: $g(\theta) = \sin(\theta) + d$

Adding d shifts the **midline** to $y = d$. The amplitude is **not affected / unchanged**.

Function	Amplitude	Max	Min	Midline
$y = \sin \theta + 3$	1	4	2	$y = 3$
$y = 2 \sin \theta - 1$	2	1	-3	$y = -1$

LO 3.6.A — Part 2: Horizontal Transformations (b and c) — Key

Horizontal Dilation: $g(\theta) = \sin(b\theta)$

If $|b| > 1$: the graph is **compressed horizontally** (period is shorter).

If $|b| < 1$: the graph is **stretched horizontally** (period is longer).

Function	b value	Period
$y = \sin(3\theta)$	3	$2\pi/3$
$y = \sin(\frac{1}{2}\theta)$	$1/2$	4π
$y = \sin(4\theta)$	4	$\pi/2$

Phase shift = $-c$ (use the factored form to read it correctly).

Important — Factoring first!

$$\sin(2\theta + \pi) = \sin(2(\theta + \pi/2))$$

Phase shift = $-\pi/2 = -\pi/2$ (**left $\pi/2$ units**)

LO 3.6.A — Part 3: The General Form — Key

General Sinusoidal Function:

$$f(\theta) = a \sin(b(\theta + c)) + d$$

Parameter	Feature	Formula
a	Amplitude	$ a $ (sign determines reflection)
b	Period	$T = \frac{2\pi}{ b }$
c	Phase shift	$-c$ (left if $c > 0$, right if $c < 0$)
d	Midline	$y = d$

Worked Example: $f(\theta) = -2 \sin(\frac{1}{2}(\theta - \pi)) + 3$

Amplitude: $|-2| = 2$

Period: $2\pi/(1/2) = 4\pi$

Phase shift: $-(-\pi) = \pi$, **shifted right π units**

Midline: $y = 3$

Reflected? **Yes, because $a = -2 < 0$**

Sine-Cosine Equivalence:

$$2 \cos \theta = 2 \sin\left(\theta + \frac{\pi}{2}\right)$$

Guided Practice

Guided Practice 1.

Function	Amplitude	Period	Phase Shift	Midline
$f(\theta) = 5 \sin(3\theta) - 2$	5	$2\pi/3$	none (0)	$y = -2$
$g(\theta) = -\cos\left(\frac{\pi}{2}\theta\right) + 4$	1	4	none (0)	$y = 4$
$h(\theta) = 3 \sin\left(2\left(\theta + \frac{\pi}{6}\right)\right)$	3	π	$-\pi/6$ (left)	$y = 0$

Guided Practice 2.

- (a) $3\theta - \pi = 3(\theta - \pi/3)$
- (b) Phase shift = $\pi/3$ **to the right**
- (c) Amplitude **4**, period $2\pi/3$, phase shift $\pi/3$ **right**, midline $y = 1$.